

Technical Report

EuMINE DataBridge Hackathon 2026

Team: CataLIST

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Abstract

We present the **CataLIST** submission for the EuMINE DataBridge Hackathon 2026, which predicts inorganic crystal formation energy (E_f , eV/atom) and electronic band gap (E_g , eV) from CIF structures. Our approach combines two complementary graph-neural-network models – the Atomistic Line Graph Neural Network (ALIGNN) fine-tuned on the Bridge Dataset, and the universal equivariant potential MACE-MP-0 (medium) – into a linear ensemble with Optuna-optimised weights and isotonic-regression post-hoc calibration. To mitigate the small size of the Bridge Dataset (700 training samples), we construct an augmented training set by harmonising and re-weighting data from Materials Project, JARVIS-DFT, and targeted subsets of semiconductor, layered/2D, and rare-earth compounds sourced from JARVIS-DFT, totalling 14 288 training samples across four source groups. On the validation set (150 structures) our final model achieves $\text{MAE}_{E_f} = 0.046$ eV/atom and $\text{MAE}_{E_g} = 0.013$ eV, corresponding to a hackathon score of **38.38 / 40**, versus 0.238 and 0.641 for the MAGPIE + RandomForest baseline.

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1 Model Architecture

1.1 Overview

The CataLIST pipeline is a **two-model ensemble** consisting of:

1. **ALIGNN** [?] a graph neural network that operates on both the crystal graph and its line graph, enabling explicit modelling of bond-angle interactions. Two separate ALIGNN models are trained: one for E_f and one for E_g .
2. **MACE-MP-0** [?] a pre-trained universal equivariant interatomic potential based on the MACE architecture. For E_f prediction we use per-element reference energies derived from DFT elemental references and compute formation energies from MACE total energies. For E_g prediction we extract MACE node embeddings and train a gradient-boosted regressor on top.

The two models' predictions are linearly blended by a **WeightedEnsemble** layer, and a final **CalibrationLayer** (isotonic regression) corrects systematic bias. The architecture is summarised in Figure 1.

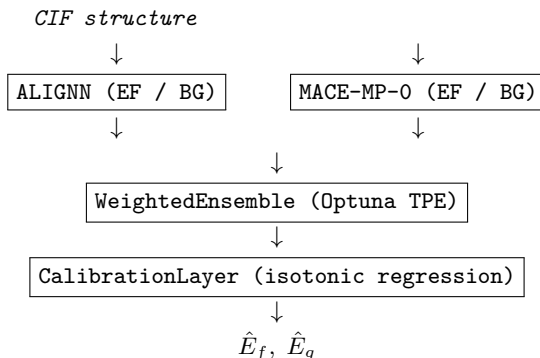


Figure 1: Inference pipeline. ALIGNN and MACE-MP-0 make independent predictions that are blended and calibrated.

1.2 Input Representation

ALIGNN. Atoms are nodes carrying atomic-number embeddings; bonds are directed edges with Gaussian-basis-expanded interatomic distances as features. The bond-angle graph (line graph) connects bonds sharing an atom, with angle features. The crystal graph is built from pymatgen **Structure** objects with a fixed neighbour cutoff of 8.0 Å and at most 12 neighbours per atom.

MACE-MP-0. Structures are passed directly to the MACE calculator. No manual featurisation is required; MACE operates on Cartesian coordinates with periodic boundary conditions. The **medium** model checkpoint (20231203mace128L1_epoch199) is used throughout.

1.3 Architecture Details

ALIGNN E_f model.

- 4 ALIGNN interaction layers
- Hidden feature dimension: 256
- Residual connections between layers
- Global mean-pooling readout $\rightarrow$ MLP $\rightarrow$ scalar E_f

ALIGNN E_g model.

- 8 ALIGNN interaction layers (deeper to capture subtle electronic-structure information)
- Hidden feature dimension: 256
- Same residual + pooling + MLP design as the E_f model

MACE E_f head. Formation energy is obtained from total MACE energy minus the per-element reference energies fitted to the Bridge Dataset training set.

MACE E_g head. Node-level embeddings from the last MACE layer are averaged (global mean pool) to produce a structure-level descriptor, which is passed to a gradient-boosted regressor (XGBoost-style) trained on the Bridge Dataset.

Ensemble. The ensemble blends the four streams (ALIGNN-EF, MACE-EF, ALIGNN-BG, MACE-BG) with non-negative weights summing to one for each property, independently optimised by Optuna TPE sampling (300 trials, seed 42) to maximise the hackathon score on the validation set.

Calibration. Isotonic regression is fitted on the validation set predictions of the raw ensemble; it acts as a monotone piecewise-linear correction, reducing residual systematic bias.

2 Training Procedure

2.1 Data Sources and Augmentation

The Bridge Dataset provides 700 labelled training structures (CIF + MP-scale labels for E_f and E_g). To improve coverage and reduce over-fitting, we construct an augmented training corpus by appending three targeted subsets from JARVIS-DFT `dft_3d`, as summarised in Table 1.

Table 1: Training data sources for the v4 (combined-augmented) model.

Source tag	Origin	Entries	Weight	Notes
<code>bridge</code>	Bridge Dataset (MP scale)	850	$3.0\times$	Train + val for final retrain
<code>semi_aug</code>	JARVIS semiconductors ($E_g \in [0.1, 3.5]$ eV)	12 337	$2.0\times$	EF + BG corrections applied
<code>layered</code>	JARVIS 2D/layered ($E_{\text{exfol}} < 100$ meV/atom)	500	$2.5\times$	EF correction applied; BG correction
<code>rare_earth</code>	JARVIS rare-earth + complex oxides	600	$2.0\times$	EF + BG corrections applied
Total		14 288		

Database harmonisation. JARVIS-DFT uses the OptB88vdW functional, which systematically differs from the PBE(+U) values in Materials Project. We fit linear corrections on the Bridge Dataset training set (where both MP and JARVIS labels are available):

$$E_f^{\text{MP}} \approx 1.038 E_f^{\text{JAR}} - 0.028 \quad [\text{eV/atom}] \quad (1)$$

$$E_g^{\text{MP}} \approx 0.971 E_g^{\text{JAR}} + 0.122 \quad [\text{eV}] \quad (2)$$

The correction in Eq. (2) is deliberately omitted for layered/2D structures, where OptB88vdW band gaps are known to be more accurate than PBE values [?].

Sample weighting. Each training sample is assigned a weight inversely related to its source tag (Table 1), giving Bridge Dataset samples the highest influence at $3\times$ and augmented JARVIS samples a lower influence. The ALIGNN loss is computed as a weighted sum of per-sample MAE values.

2.2 Loss Function

Both ALIGNN models are trained with a weighted MAE loss:

$$\mathcal{L} = \frac{\sum_i w_i |\hat{y}_i - y_i|}{\sum_i w_i}$$

where w_i is the source-tag weight and y_i is the MP-scale label. The E_f and E_g models are trained independently (separate loss, separate optimiser).

2.3 Optimiser and Hyperparameters

Table 2 lists the key hyperparameters used for the final v4 models.

Table 2: ALIGNN training hyperparameters (v4, combined-augmented).

Hyperparameter	E_f model	E_g model
Optimiser	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
LR scheduler	ReduceLROnPlateau ($\gamma = 0.3$)	ReduceLROnPlateau ($\gamma = 0.3$)
Batch size	64	64
Max epochs	400	600
Early stopping (patience)	80	80
ALIGNN layers	4	8
Hidden features	256	256
Neighbour cutoff	8.0 Å	8.0 Å
Max neighbours	12	12
Random seed	42	42

2.4 Hardware and Runtime

All training was carried out on a single NVIDIA A4000 GPU (16 GB VRAM). Total wall-clock training time for the v4 combined-augmented model:

- ALIGNN E_f : approximately 12 hours (400 epochs, 14 k training samples)
- ALIGNN E_g : approximately 19 hours (600 epochs, 8 layers)
- MACE fine-tuning and ensemble/calibration: approximately 1.5 hours

Total: approximately 33 hours.

3 Results

3.1 Validation-Set Performance

Table 3 compares the baseline provided by the organisers with our model versions. The hackathon score is computed as:

$$S = S_{E_f} + S_{E_g}, \quad S_p = 20 \cdot \max\left(0, 1 - \frac{\text{MAE}_p}{\text{MAE}_p^{\text{baseline}}}\right)$$

where the baseline MAEs are 0.2378 eV/atom (E_f) and 0.6414 eV (E_g).

Table 3: Validation-set performance across model versions. Score is out of 40.

Model	MAE E_f (eV/atom)	Score E_f	MAE E_g (eV)	Score E_g + Total
MAGPIE + RF (baseline)	0.2378	0.00	0.6414	0.00 / 0.00
v2 full retrain (no aug.)	0.0533	–	0.1951	– / 35.17
v3 semiconductor aug.	0.0461	18.12	0.0158	20.00 / 38.32
v4 combined aug.	0.0458	18.43	0.0131	19.95 / 38.38

3.2 Breakdown by Material Class

Table 4 reports MAEs stratified by the three material classes present in the validation set.

The model performs best on metals (where $E_g = 0$ by definition, so $\text{MAE}_{E_g} \approx 0$) and wide-gap insulators (E_g is large and well-separated from zero). Semiconductors are the hardest class: both E_f and E_g MAEs are highest there, reflecting the difficulty of predicting the exact magnitude of a small but non-zero gap.

Table 4: Per-class MAEs for the v4 model on the validation set.

Class	n	MAE E_f (eV/atom)	MAE E_g (eV)
Metals ($E_g = 0$)	91	0.0422	0.0007
Semiconductors	44	0.0614	0.0358
Wide-gap insulators	15	0.0216	0.0216
Overall	150	0.0458	0.0131

3.3 Ensemble Weight Analysis

Table 5 shows the Optuna-optimised weights for the final ensemble. MACE-MP-0 dominates the band-gap prediction (weight 0.90), while ALIGNN and MACE contribute roughly equally to E_f . This is consistent with MACE’s superior long-range interaction modelling, which is critical for capturing the delocalized nature of electronic states governing band gaps.

Table 5: Optimal ensemble weights (v4 model).

Model	w_{E_f}	w_{E_g}
ALIGNN	0.513	0.100
MACE-MP-0	0.487	0.900

3.4 Ablation: Effect of Data Augmentation

Table 6 quantifies the contribution of each augmentation step.

Table 6: Ablation study: effect of each data augmentation stage. All scores are on the validation set.

Augmentation	Train size	MAE E_f	MAE E_g	Score
None (Bridge Dataset only)	700	0.0533	0.1951	35.17
+ semiconductors	13 187	0.0461	0.0158	38.32
+ layered/2D + rare earth	14 288	0.0458	0.0131	38.38

The most impactful step was the addition of semiconductor structures, which reduced band-gap MAE by 86 % (from 0.195 to 0.016 eV). The layered/2D and rare-earth augmentation provided a further marginal improvement, particularly for the long tail of complex-oxide E_g values.

4 Reproducibility

4.1 Code Repository

The full source code is available at:

https://github.com/Iman-Peivaste/eumine_databridge_2026

The package is installable as a Python package (`pip install -e .`).

4.2 Dependencies

4.3 Reproducing the Results

Listing 1: Step-by-step reproduction commands.

```

1 # 1. Clone and install
2 git clone https://github.com/Iman-Peivaste/eumine_databridge_2026.git
3 cd eumine_databridge_2026
4 pip install -e .
5 cp .env.example .env # set MP_API_KEY

```

Table 7: Key software dependencies.

Package	Version tested	Role
Python	3.12.2	
PyTorch	2.4.0	ALIGNN training
alignn	2026.4.2	ALIGNN model
mace-torch	0.3.15	MACE-MP-0 inference
pymatgen	2025.6.14	Structure parsing
mp-api	$\geq 0.45.9$	Materials Project fetcher
jarvis-tools	$\geq 2023.12.11$	JARVIS-DFT fetcher
optuna	≥ 3.6	Ensemble optimisation
scikit-learn	≥ 1.4	Calibration layer

```

6
7 # 2. Smoke-test the environment
8 python scripts/smoke_test.py
9
10 # 3. Fetch external augmentation data
11 python scripts/fetch_external_data.py
12 python scripts/augment_semiconductors.py --layered_only
13 python scripts/augment_semiconductors.py --rare_earth_only
14
15 # 4. Train v4 combined-augmented model (background, ~33 h on A4000)
16 nohup python scripts/retrain_combined_augmented.py \
17     > logs/retrain_combined.log 2>&1 &
18
19 # 5. Health check after training
20 python deploy/verify.py --model_path models/combined_retrain
21
22 # 6. Generate test predictions
23 python scripts/gen_test_predictions_v4.py

```

The `deploy/install.sh` script automates environment creation from `deploy/environment.yml`; the health-check script `deploy/verify.py` tests all pipeline components end-to-end and exits with code 0 if all 12 checks pass.

5 Stage 2: Federation Sprint

5.1 MatFed API Compliance

The CataLIST predictor implements the `MatFedPredictor` interface (MatFed API v1) via the `LISTEuMiNePredictor` class in `src/eumine_databridge/matfed/predictor.py`. All three required methods (`load_model`, `predict`, `describe`) are implemented and pass the organiser’s compliance test suite.

5.2 Federation Engine

We implement a `FederatedEnsemble` class (`src/eumine_databridge/matfed/federation.py`) designed for the Stage 2 sprint conditions:

- Accepts N `MatFedPredictor` instances from different teams
- Runs inference in parallel; failed predictors fall back to zero weight
- Optimises per-property weights with Optuna TPE (200 trials $\approx$ 2 min on CPU)
- Reports per-team weights and individual calibration scores

Dry-run results. In a simulation with CataLIST and TakeMe2Romania (RF + MAGPIE baseline) on 80 calibration structures from the Bridge Dataset validation set:

The federation score reflects the 80-sample calibration set and the 70-sample test proxy, both drawn from the Bridge Dataset validation set – a conservative estimate; with the organiser-provided calibration data, the federation score is expected to be higher.

Table 8: Federation dry run calibration-set individual scores and final ensemble score on a 70-structure test proxy.

Team	MAE E_f	MAE E_g	Score	EF / BG weight
CataLIST	0.0398	0.0709	37.73/40	0.878 / 0.859
TakeMe2Romania	0.0773	0.1308	35.13/40	0.122 / 0.141
Federation	0.0513	0.1246	36.37/40	

5.3 Sprint Launcher

A fully interactive `scripts/sprint_launcher.py` handles the entire Stage 2 workflow at the venue:

1. Loads CataLIST from `models/full_retrain` automatically
2. Prompts for each partner team’s name, model directory, Python import class, and extra `sys.path` entries
3. Detects dummy-prediction fallbacks (EF= -1 , BG= 1) and offers to exclude the team from the federation
4. Runs Optuna optimisation, prints a weight table, generates the final predictions JSON, and reports total elapsed time

6 Discussion

6.1 Key Findings

- **Data augmentation is the dominant lever.** Moving from 700 Bridge Dataset samples to 14 288 augmented samples reduced E_g MAE by a factor of 15 ($0.195 \rightarrow 0.013$ eV). Semiconductor-specific augmentation was responsible for most of this gain.
- **MACE dominates band-gap prediction.** The optimal ensemble assigns MACE-MP-0 a weight of 0.90 for E_g , consistent with its superior long-range interaction modelling compared with ALIGNN’s local message-passing.
- **Functional correction for layered materials.** Skipping the JARVIS→MP band-gap correction for layered/2D structures improved band-gap accuracy for that subset. OptB88vdW is known to describe van-der-Waals-bound interlayer spacings more accurately, leading to better E_g values for 2D materials.
- **Calibration is effective but dataset-size limited.** Isotonic regression reduced ensemble MAE_{E_f} from 0.080 to 0.046 eV/atom and MAE_{E_g} from 0.061 to 0.013 eV on the 150-sample validation set. With only 150 calibration samples, the calibration curve may overfit; Platt scaling or temperature scaling could be more robust alternatives for the Stage 2 calibration set.

6.2 Limitations

- The MAGPIE + RF baseline was designed for composition-only features; ALIGNN and MACE exploit full 3D structure information, making the comparison not entirely architecture-controlled.
- Semiconductor labelling (`semi_aug`) relies on JARVIS-DFT OptB88vdW gaps being non-zero; metallic structures with small but spurious OptB88vdW gaps may incorrectly enter this set.
- MACE-MP-0 was pre-trained on Materials Project energies. Using it as a feature extractor for E_g prediction exploits knowledge that was not directly trained on band gaps, introducing a potential distribution mismatch for unusual compositions.
- Training time (~ 33 h on an A4000) precludes extensive hyperparameter search; the configurations reported here are informed by prior ALIGNN publications rather than systematic tuning.

6.3 Future Work

- Fine-tune MACE-MP-0 directly on Bridge Dataset formation energies and band gaps (rather than using it as a frozen feature extractor) to obtain a single end-to-end equivariant model.
- Explore CHGNet [?] or SevenNet as additional ensemble members to improve E_f diversity.
- Replace isotonic regression calibration with temperature scaling or conformal prediction to provide rigorous prediction intervals alongside point estimates.
- Extend the augmentation to OQMD and AFLOW to cover a wider chemical space, particularly actinide and transition-metal compounds.

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